

Sentiment-aware Classifier for Out-of-Context Caption Detection

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ABSTRACT

In this work we propose additions to the COSMOS and COSMOS on Steroids pipelines for the detection of Cheapfakes for Task 1 of the ACM Grand Challenge for Detecting Cheapfakes. We compute sentiment features, namely polarity and subjectivity, using the news image captions. Multiple logistic regression results show that these sentiment features are significant in prediction of the outcome. We then combine the sentiment features with the four image-text features obtained in the aforementioned previous works to train an MLP. This classifies sets of inputs into being *out-of-cotext* (OOC) or *not-out-of-context* (NOOC). On a test set of 400 samples, the MLP with all features achieved a score of 87.25%, and that with only the image-text features a score of 88%. In addition to the challenge requirements, we also propose a separate pipeline to automatically construct caption pairs and annotations using the images and captions provided in the large, un-annotated training dataset. We hope that this endeavor will open the door for improvements, since hand-annotating cheapfake labels is time-consuming. To evaluate the performance on the test set, the Docker image with the models is available at: <https://hub.docker.com/repository/docker/malkaddour/mmsys22cheapfakes>. The open-source code for the project is accessible at: <https://github.com/malkaddour/ACMM-22-Cheapfake-Detection-Sentiment-aware-Classifier-for-Out-of-Context-Caption-Detection>.

CCS CONCEPTS

• Computing methodologies → Neural networks;

KEYWORDS

Cheapfake detection, COSMOS, sentiment analysis, NLP

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1 INTRODUCTION

The detection of cheapfakes in news setting is useful in the combatting misinformation. Cheapfakes are images or text that have been manually altered to propagate a false narrative. The seminal COSMOS dataset produced by Aneja *et al.* [3] is the first of its kind to group a large number of news with their captions and construct a test set containing cheapfakes in the form of captions. The dataset is scrapped from online news sources and contains images and their metadata. The article topics in which the images are retrieved from are those that are prone to misinformation [3]. While some captions are descriptive and objective, journalists also choose to write opinion-laden captions that suggest the author's sentiment towards the image.

To the best of our knowledge, sentiment-aware classifiers for out-of-context detection does not exist in the literature. Due to the importance of sentiment analysis in news articles in general, we hypothesize that incorporating sentiment into context-detection could help improve the classification accuracy for this task.

We propose in this work to contextualize the sentiment of the captions during classification. Specifically, we propose to compute sentiment features and fuse them with the baseline features from the 2021 Cheapfakes Detection winning research, COSMOS on Steroids [2], to create a sentiment-aware classifier for out-of-context detection in this setting. We also train a multi-layer perceptron at the pipeline output to test how well a deep network can learn the features for cheapfakes detection.

2 RELATED WORK

2.1 Image-text analysis

As seen in the baseline COSMOS [3] and COSMOS on Steroids [2], the image-text matching task yields key features that aid in context classification. In their works, they determine which objects the image captions refer to and use this information to aid in out-of-context detection for caption pairs.

Bao *et al.*'s VLMO [16] can learn advanced image and text embeddings for image-text pairs and have achieved record numbers on both Visual Question Answering (VQA) and Natural Language for Visual Reasoning 2 (NVLR2) datasets. Other high-performing architectures for visual-textual tasks are COCA by Yu & Wang [22] and Neural Symbolic VQA by Yi *et al.* [21].

Captioning of news images is another image-text task highly related to this work. Tran, Mathews, and Xie [14] introduce the *Transform-and-tell* image captioner, which relies on the entirety of the news article as well as the location of the image in the article to automatically generate realistic news captions. A similar work is also done by Liu *et al.* [7] using *VisualNews* to achieve improved results on captioning datasets. More general image captioners are also available, such as [8].

2.2 Sentiment analysis

Sentiment analysis is the process of extracting information about the writers' emotional involvement in the text. Two commonly used metrics are polarity and subjectivity. Polarity assesses whether the sentiment is positive, neutral, or negative, while subjectivity determines how objective the statement is.

One of the most popular sentiment analyzers is the *Valence Aware Dictionary and sEntiment Reasoner* (VADER) by Hutto & Gilbert [5], which uses a rule-based approach to predict text polarity. It performs well on contexts such as social media, news articles, reviews, and blogs. Recent sentiment analysis methods have also been explored. Some of the methods propose deep learning-based methods for sentiment analysis, such as by Abid *et al.* [1] and Yang *et al.* [20]. Wankhade *et al.* [17] propose the *context-based vectors over sentiment domain*, or CBVoSD, method, in which they incorporate a word sentiment vector into their model for the task of sentiment classification on reviews. The interested reader can consult the review papers by Wankhade *et al.* [18], Ligthart *et al.* [6], and Parvin *et al.* [10] for a more complete overview of the advancements in sentiment analysis.

Some other works are involved specifically in analyzing sentiments of full news articles. Yadav *et al.* [19] focus on sentiment analysis for financial news specifically, which they do by building on an unsupervised method by Turney [15]. Some other works on news sentiment analysis include [12], [13], and [11].

3 PROPOSED METHOD

Our method consists of three main processes: preliminary regression on the sentiment features, COSMOS training data annotation, and finally the fusion of the sentiment features with the features obtained from the baseline COSMOS on Steroids [2] using a multilayer perceptron (MLP) as a binary classifier. The preliminary regression is implemented to gain insight on the predictive capacity of the sentiment features that we compute on the given triplet, that consists of two input captions and one input image. Following that, the appropriate function to best classify the OOC/NOOC binary label is learned using an MLP with inputs consisting of both the sentiment features and the features used in the state-of-the-art Cosmos on Steroids work [2].

3.1 Sentiment features

To analyze the sentiments present in the captions, two sentiment features are extracted for a given caption: polarity p and subjectivity s . For $p \in [-1, 1]$, values such that $|p| < t_p$ are considered neutral, while those above and below t_p are positive and negative sentiments, respectively. Here, t_p is a threshold chosen to determine

the neutral range for polarity. For $s \in [0, 1]$, $s = 1$ is considered highly subjective, while $s = 0$ is considered highly objective.

The state-of-the-art VADER [5] was leveraged to compute the polarity values for a given caption, in which the neutrality threshold is $t_p = 0.05$. Subjectivity values are computed using *TextBlob*, which is an open-source natural language processing package available in Python [9].

For our preliminary analysis, we use the publicly available annotated test set from COSMOS consisting of 1000 images, each with a caption pair $C_i = \{C_i^1, C_i^2\}$ and a binary ground-truth label y_i , where $y_i = 0$ and $y_i = 1$ refer to NOOC and OOC, respectively. C_i^1 and C_i^2 are the full caption texts obtained from the dataset. These features are distinctive from the features previously used in [3] and [2], which were image-to-text matching and sentence similarity. If the features were found to be significant in the preliminary analysis, then they would be useful inputs to the overall classifier. We use Scikit-learn's logistic regression module to carry out this analysis using 850 annotated samples from the test set. The two features used in logistic regression are the sentiment differences between the two captions, which are $x^1 = |p^1 - p^2|$ and $x^2 = |s^1 - s^2|$.

3.2 Data annotation

Next, we use the available training set of 160k images, each with captions from multiple articles, to generate a dataset annotated with OOC/NOOC labels. We use this annotated dataset to train the MLP with all the features and to test on the manually annotated dataset of 1000 samples [3].

3.2.1 Obtaining the second caption. Let $\{(I_i, C_i^1)\}_{i=0}^N$ denote the training set with N samples, consisting of images and their associated caption. While several captions are available per image, we use one randomly selected caption in this work. For a given pair (I_i, C_i^1) , an NOOC label is assigned by paraphrasing C_i^1 to obtain ${}^0C_i^2$ (the left superscript refers to NOOC and OOC second caption, 0 for NOOC and 1 for OOC). The *punkt* model from the Natural Language ToolKit (NLTK) [4] is used for tokenization. The tokens are then used as input in the *Pre-training with Extracted Gap-sentences for Abstractive Summarization* (PEGASUS) transformer [23] to paraphrase the sentences.

We obtain five different paraphrased versions of the original captions, and select the version that is most sentimentally different than the original caption to ensure it is sufficiently different. The decision for ${}^0C_i^2$ is then simply the caption that maximizes the quantity $(p_j - p_1)^2 + (s_j - s_1)^2$, $j = 2, \dots, 6$, where $j = 1$ refers to the original caption values. The p_j and s_j are respectively the polarity and sentiment values computed for each paraphrased caption.

To obtain the OOC second caption ${}^1C_i^2$, we sample a caption from another image in the dataset. An index α_i that points to the second image is uniformly randomly sampled from the set $S_i = \{0, \dots, N\} \setminus \{0, \dots, i\} \cup \{\alpha_j\}_{j=0}^{i-1}$. In other words, S_i contains all the indices except for those that have been used previously to obtain OOC labels. Thus, the dataset contains $N + \lfloor \frac{N}{2} \rfloor$ labeled triplets (I, C^1, C^2) , with N NOOC and $\lfloor \frac{N}{2} \rfloor$ OOC labels.

3.2.2 Computing the features. The feature vector $\mathbf{x}_i \in \mathbb{R}^6$ for a given caption pair (I_i, C_i^1, C_i^2) contains both sentiment and features from COSMOS on Steroids. x_i^1, x_i^2 are the polarity and subjectivity

differences between the two captions, obtained as described in Section 3.1. The remaining four features are the textual similarity S_{sim}^i between the two captions, the Intersection over Union (IoU) of the bounding boxes, and the two binary variables which are a result of their differential sensing and fake-or-fact routines [2]. These four features are obtained by using the publicly available source code provided by [2], minus the thresholding decision process they implement, since we already have the OOC/NOOC labels as described in 3.2.1.

To compute the bounding boxes, the COSMOS on Steroids implementation requires "modified captions", which are the captions with entities' proper nouns replaced by their generic name (e.g. "Facebook" is replaced by "ORG"). To avoid the computational cost associated with passing our paraphrased caption through Spacy's Entity Recognizer (ER), we use the entity list already provided by COSMOS [3] on the original caption and replace any occurrences of the entities in our paraphrased caption. Note that we do not use the modified captions when we compute the sentiment features. The overall procedure, along with the MLP, is shown in Figure 1.¹ Together, the aforementioned procedures yield the six-dimensional feature vector $\mathbf{x}_i$, the bounding box coordinates, and the binary OOC/NOOC label y_i for a given image and caption pair.

3.3 MLP for out-of-context classification

We use an MLP with the feature vector $\mathbf{x} \in \mathbb{R}^6$ defined in Section 3.2.2 and a binary output. The neural network consists of ten fully connected layers with 1000 nodes per layer. ReLU activation is used after each hidden layer, sigmoid activation with one node for the output layer, and a dropout of 0.5 is used to avoid overfitting on the training data.

The objective function $\mathcal{L}$ used is the binary cross-entropy loss, given by

$$\mathcal{L}(\mathbf{X}, \mathbf{y}, \hat{\mathbf{y}}) = -\frac{1}{N} \sum_i^N y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i), \quad (1)$$

minimized using Adam with learning rate $\alpha = 10^{-4}$ and $\beta_1 = 0.9, \beta_2 = 0.999$. The $y_i, \hat{y}_i$ are respectively the ground-truth and predicted outputs. We varied the number of epochs between 300-800 depending on the experiment.

Two experiments were conducted using the aforementioned MLP. The first consists of training the data on a validation set which we carve out of our generated labels and testing it on both a separate subset of the same dataset and the publicly annotated test set. The second experiment is the same, except the training data is taken from a subset of the publicly annotated set instead. We remark that the subset used in the validation set contains an equal number of OOC and NOOC labels, despite the 2:1 class imbalance present in the full dataset due to our approach.

Furthermore, the MLP trained on 600 samples of the publicly annotated test set is also evaluated on the remaining 400 samples using MLPs with various combinations of input features. Specifically, we try this with all six features, only COSMOS on Steroids features, and polarity + COSMOS on Steroids features.

We do the above in three ways: 1) MLP classifies the outcome directly, 2) MLP classifies the outcome if the conditions for out-of-context as done in [2] are not met, and 3) thresholding only without MLP. In their work [2], an NOOC label is automatically given when those conditions are not met, while we allow the MLP to decide when the conditions are not met. For the cases in which we use sentiment features, our thresholding is defined by classifying the triplet as OOC when the quantity $|p^1 - p^2|^2 + |s^1 - s^2|^2 > t_s$, whether or not the original conditions for OOC are met. This is set to catch caption pairs that are in fact OOC but were not caught by the original conditions.

4 RESULTS AND DISCUSSION

4.1 Logistic regression results

A multiple logistic regression was performed on all 1000 samples of the annotated test set, yielding the model

$$\hat{y}_i = g(-0.6987 + 1.5047x_i^1 + 1.0841x_i^2), \quad (2)$$

where $g(\cdot)$ is the sigmoid function and $x_i^1 = |p_i^1 - p_i^2|$, $x_i^2 = |s_i^1 - s_i^2|$ denote the polarity and subjectivity difference between the two captions. The coefficients align with our intuition; an increase in both polarity and subjectivity difference between the two captions will increase the probability of an OOC label. The polarity difference coefficient and z-value (1.5047, 6.424) are larger than the subjectivity difference coefficient and z-value (1.0841, 4.156), indicating that the polarity difference is a stronger influencer on classification than subjectivity difference. Moreover, $x^1 \in [0, 2]$ while $x^2 \in [0, 1]$, meaning polarity difference has more potential to influence the outcome than subjectivity difference.

All three coefficients are statistically significant at the 1% confidence level with z-values of -6.736, 6.424, and 4.156 for $\hat{\beta}_i$ in that order. However, we cannot rely on Equation (2) to accurately model the relation between sentiment features and context label, but only to explore the predictive capacity of the features are reasonable inputs to a classifier.

4.2 Out-of-context classification results

The results for the binary classification using the MLP and training details described in section 3.3 are presented in Table 1. The highest accuracy achieved when tested on the dataset generated in this work ("Validation") is 97.44%. The same model achieves a score of 76% on the publicly available test set ("Test"), which is due to the difference in both a) the second caption, C_2 , and b) the expertise involved in hand-crafting the labels in Test, which used human knowledge as opposed to the automated approach of the generated validation set. We denote the subset of Test used for training by Test600, and the subset containing the remaining samples by Test400. The network trained on Test600 achieved 87.25% on Test400, which is close to the record baseline score [2] despite being tested on a possibly different subset.

Based on Table 1, the best-performing architecture was used with different combinations of the features to determine those that best contributed to out-of-context detection. These results are shown in Table 2. When all features are used, we see that the combination of the MLP with the thresholding as explained in Section 3.3 is the best method. When only the four image-text features are used,

¹The code used for this work is available at <https://github.com/malkaddour/ACMM-22-Cheapfake-Detection-Sentiment-aware-Classifier-for-Out-of-Context-Caption-Detection>.

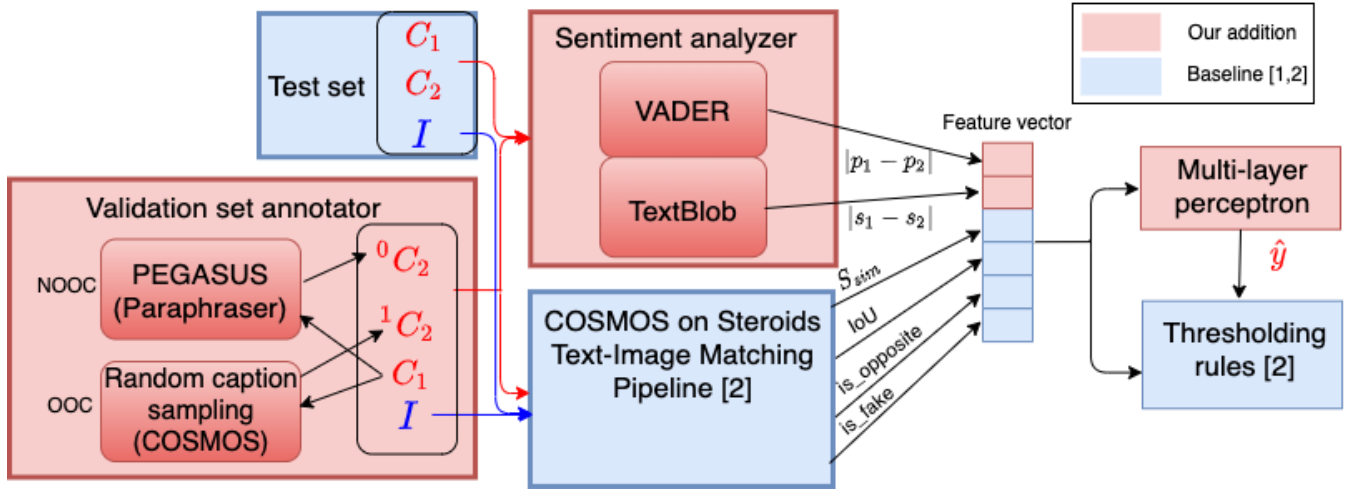


Figure 1: Overall diagram demonstrating our method, including annotation of the validation set and the inclusion of the MLP in decision-making. Note that in this work, the caption pair may come from either the validation set or test set on the left.

Table 1: Accuracy of MLP classifier. Test is the manually annotated test set by [3], and Validation is their training set with our generated data. The values next to the data represents the number of samples used.

Trained on	Tested on			
	Validation17.7k	Test1000	Validation1000+Test1000	Test400
Validation41k	97.44%	77.75%	91.50%	76%
Test600	76.01%	—	78.50%	87.25%

the performance is improved and the threshold-only method is superior. For a test set of 400 samples, this correspond to two extra samples correctly classified. When polarity is added to the image-text features, the MLP+threshold is again the highest with a score of 87.50%, and the threshold only method is lowest at 68.25%.

These results indicate that for the test dataset, using the technique proposed by [2] remains marginally superior to the modifications in this work. However, we also observe that an MLP-based training approach on the features and ground-truth labels without any thresholding also yields roughly equivalent and competitive scores using different feature combinations, while the thresholding method on its own depends more on the choice of features used. Hence, for the 2022 challenge, we choose the MLP trained on the full 1000 samples of the test dataset using all six features due to its robustness and potential to score well without handcrafted rules.² Further, analysis on a separate dataset will be required as future work to confirm this observation.

5 CONCLUSION

In this work, we implement a procedure to incorporate sentiment features that are associated with news captions to assist in the task of cheapfake detection. Polarity and subjectivity values are

Table 2: Accuracy of MLP classifier trained on Test600 using different combinations of features. Image-text refers to all four features computed by [2].

Feature combination	MLP only	MLP + threshold	threshold only
All features	87.25%	87.50%	72.50%
Image-text only	88.00%	88.25%	88.50%
Polarity + Image-text	87.50%	87.75%	68.25%

extracted from image pairs using VADER [5] and TextBlob [9] respectively to be used as features in an MLP to predict NOOC/OOC labels. These features are combined with the state-of-the-art COSMOS on Steroids model [2].

Performance is tested using a validation set that we automatically annotated from the COSMOS training set [3] as well as their handcrafted test set. The results show that the MLP trained on the handcrafted set using the features in the prior work and the sentiment features in this work can perform well on the test set at the 87.25%. Although this does not surpass the performance of COSMOS on Steroids [2], it demonstrates the potential of leveraging a well-trained MLP for this task without a thresholding protocol, and is a baseline for future work in this direction.

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²The Docker image containing the model can be accessed at <https://hub.docker.com/repository/docker/malkaddour/mmsys22cheapfakes>

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